Supplementary Material: Sparse Feature Selection, Not Capacity, Drives Spatial Transfer in Crop Classification

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This document collects supporting material for the main paper. Section S-A details the dataset, feature set, and model definitions; Section S-B defines the metrics; Section S-C reports training cost; and Section S-D presents the ablation studies. Equation, figure, table, and section labels are prefixed "S".

S-A. Dataset Details, Feature Set, and Model Definitions

Supports main paper Sec. 2–3.

S-A.1. Exploratory Data Analysis

The five crop classes are heavily imbalanced (Fig. S1(a)). Lucerne/medics dominates both regions despite having the smallest mean field area (Fig. S1(b)). Spectral responses overlap substantially across crops (Fig. S2(a)), and wheat and barley are especially similar (Fig. S2(b)), the dominant source of persistent confusion noted in the main paper.

S-A.2. `xr_fresh` Feature Set

Table S1 lists the time-series features extracted from each pixel's 10-month spectral profile with `xr_fresh` (Mann et al., 2026; Mann, 2025), forming the 174-dimensional per-pixel vector used by the classical learners (Fig. S3 illustrates the extraction).

S-A.3. Baseline Classical Model Definitions

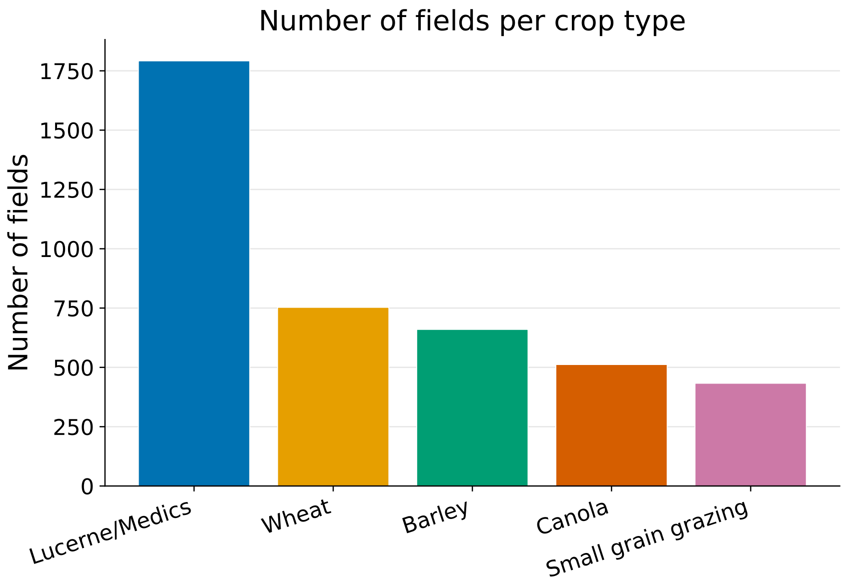
For a feature vector $\mathbf{x}_i$ and class $c \in \{1, \dots, C\}$, the four baselines are as follows. Logistic regression (James et al., 2013) (multinomial):

$$P(y_i = c \mid \mathbf{x}_i) = \frac{\exp(\mathbf{w}_c^\top \mathbf{x}_i + b_c)}{\sum_{c'=1}^{C} \exp(\mathbf{w}_{c'}^\top \mathbf{x}_i + b_{c'})}. \tag{S1}$$

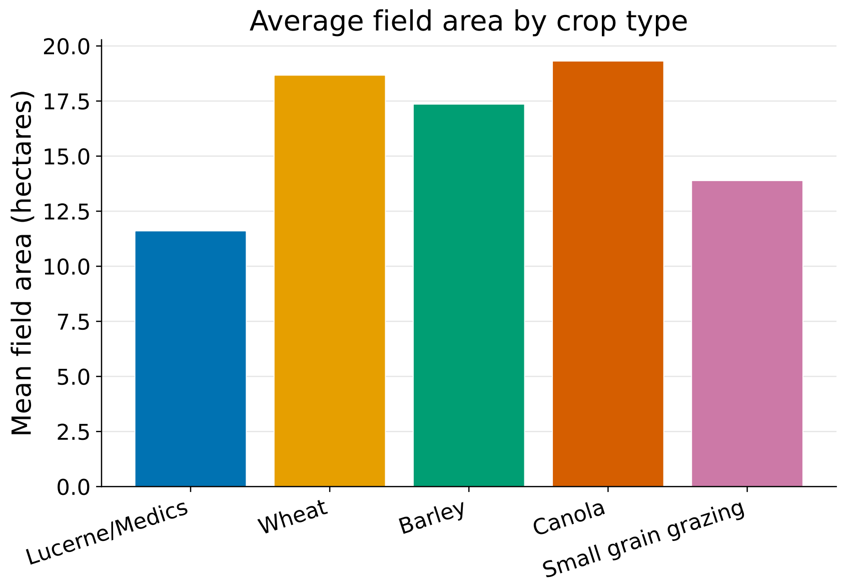
Random Forest (Breiman, 2001): majority vote over K trees, $\hat{y}_i = \mathrm{mode}\{T_1(\mathbf{x}_i), \dots, T_K(\mathbf{x}_i)\}$. LightGBM (Ke et al., 2017): additive boosting with rate η, $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta\, f_t(\mathbf{x}_i)$. XGBoost (Chen and Guestrin, 2016; Friedman, 2001): regularized boosting objective $\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$ with $\Omega(f_k) = \gamma T_k + \frac{1}{2}\lambda \sum_j w_j^2$. The field-level XGBoost was tuned with Optuna over 150 trials; the top configurations appear in Table S2. Field-level pixel features are aggregated by the mean within each polygon. For the pixel ensemble, RF, XGBoost, and LightGBM probabilities are soft-voted; the field-level stacked ensemble uses RF and XGBoost base learners with a logistic-regression meta-learner, and a confidence-margin rule falls back to pixel-mode voting when the top two field-level class probabilities differ by less than 0.10.

S-A.4. Deep Model Definitions

CNN-BiLSTM. An input tensor of shape $(B, 6, T)$ passes a 1D convolution (64 filters, kernel 5) and a bidirectional LSTM (hidden size 64), with the final timestep classified via a softmax head (Fig. S4). Class imbalance is handled with focal loss $\mathcal{L}_{\text{focal}} = -\alpha_t(1 - p_t)^\gamma \log(p_t)$, $\gamma = 2.0$, α_t the inverse class frequency. Five seeds are ensembled by averaging logits.

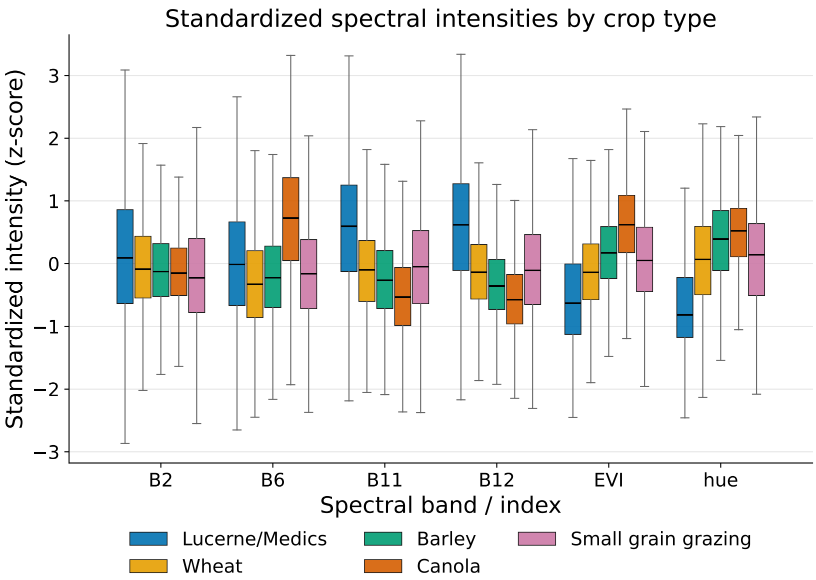


(a)

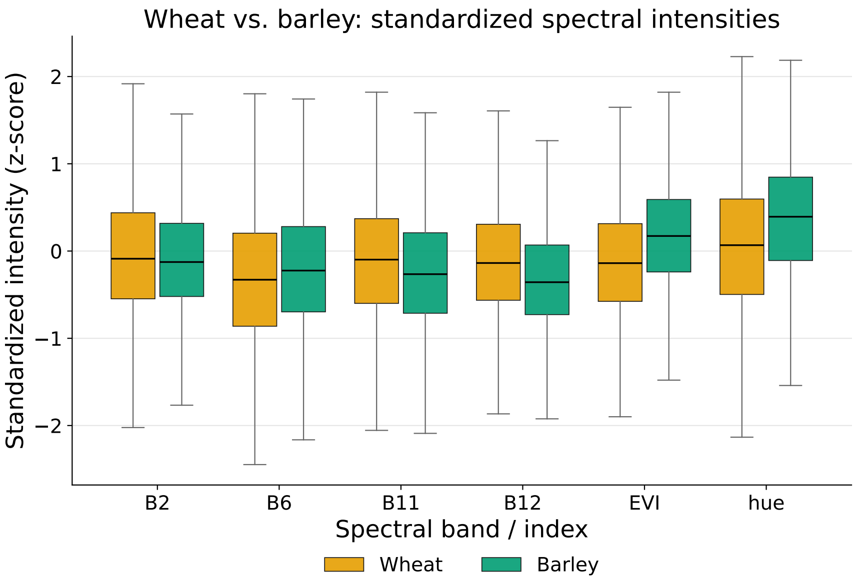


(b)

Figure S1: Crop-field composition of the training and holdout regions. (a) Number of labeled fields per crop type: lucerne/medics dominates (43% of training and 56% of holdout fields), followed by wheat, barley, canola, and small-grain grazing. This heavy class imbalance motivates macro-F1 as the primary metric and the class-imbalance handling used during training. (b) Mean field area by crop type (hectares); lucerne/medics is the most frequent class despite having among the smallest fields.



(a)



(b)

Figure S2: Spectral separability of the crop classes, shown as box plots of standardized (z-score) intensity for each Sentinel-2 band and index (B2, B6, B11, B12, EVI, hue). (a) Distributions for all five crops overlap substantially across every band, so no single feature cleanly separates the classes. (b) Wheat versus barley are especially similar in every band—the dominant source of the persistent wheat–barley confusion seen across all models in the main paper.

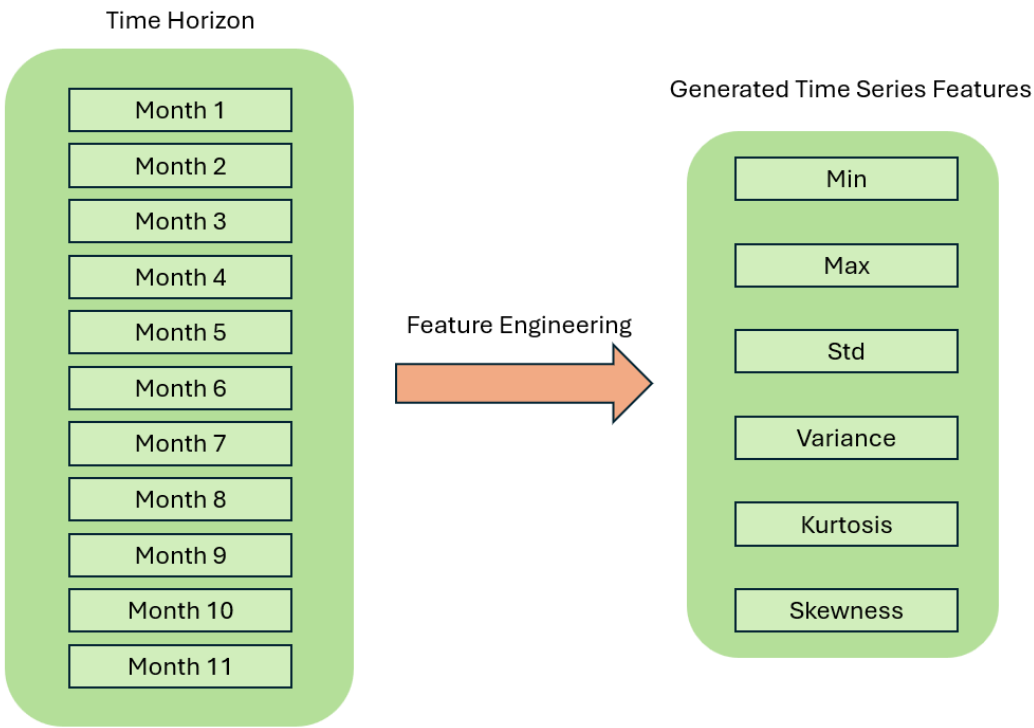


Figure S3: Time-series feature generation with `xr_fresh`. For each pixel, the 10-month profile of every band and index is summarized by the statistical and temporal descriptors of Table S1 (e.g., linear trend, autocorrelation, quantiles, day-of-year of the maximum), producing the 174-dimensional per-pixel feature vector used by the classical learners.

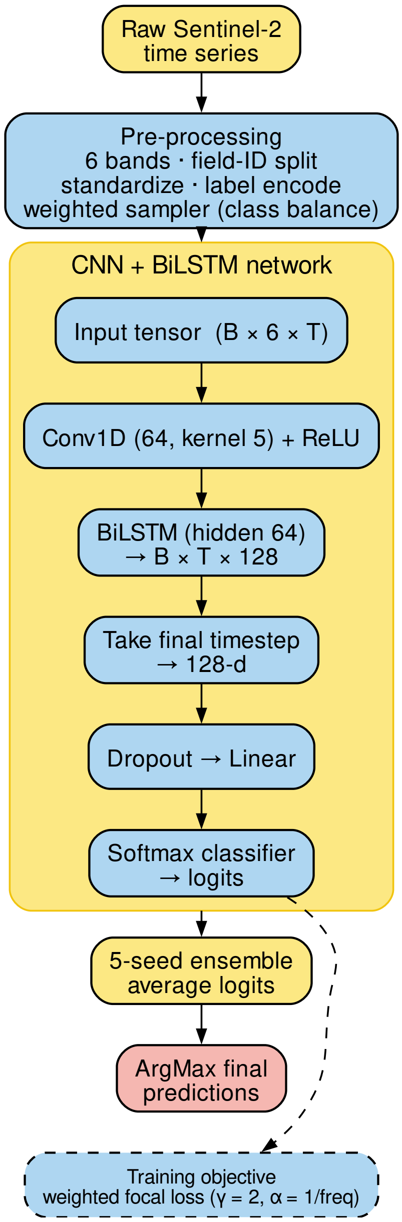


Figure S4: Architecture of the CNN-BiLSTM model. The $(B, 6, T)$ input (six bands/indices over T months) passes a 1D convolution (64 filters, kernel 5) for temporal feature extraction, then a bidirectional LSTM (hidden size 64) whose final time step is classified by a softmax head. Training uses weighted focal loss to counter class imbalance, and five seeds are ensembled by averaging logits.

Table S1: Time-series features extracted from each pixel's spectral profile with `xr_fresh`.

Feature	**Description**
Absolute Energy	Sum of squared values.
Absolute Sum of Changes	Total absolute difference between consecutive measurements.
Autocorrelation (1, 2-month lag)	Correlation with one- and two-month lagged versions.
Count Above Mean	Number of points above the series mean.
Day of Year of Max/Min	Julian day of the maximum and minimum.
Kurtosis	Tail heaviness of the distribution.
Linear Trend	Slope of a least-squares fit.
Longest Strike Above/Below Mean	Longest consecutive run above or below the mean.
Max/Min	Absolute extreme values.
Mean/Median	Measures of central tendency.
Mean Absolute Change/Mean Change	Average magnitude and signed change between points.
Mean Second Derivative	Mean of the series' second differences.
Quantiles (0.05, 0.95)	5th and 95th percentiles.
Ratio Beyond $r\sigma$	Proportion of points more than r s.d. ($r = 1, 2, 3$) from the mean.
Skewness	Asymmetry around the mean.
Standard Deviation/Variance	Spread measures.
Sum of Values	Total sum of observations.
Symmetry Looking	Similarity under series reversal.
Time Series Complexity	Complexity-Invariant Distance metric of irregularity.

Each descriptor is computed independently for every band and index from a pixel's 10-month series; stacked across the six bands/indices (B2, B6, B11, B12, EVI, hue) they form the 174-dimensional per-pixel vector fed to the classical learners. For the Ratio-Beyond features, r is the threshold in standard deviations ($r = 1, 2, 3$).

Table S2: XGBoost hyperparameter search: top 5 Optuna trials with validation accuracy.

Trial	**Acc.**	**Est.**	**Depth**	**LR**	**SubS**	**ByTree**	**Gamma**	**Regα**	**Regλ**	**MinCW**
32	**76.61%**	1141	13	0.0215	0.5749	0.6986	0.5589	0.00184	0.00259	5
12	76.46%	984	12	0.0145	0.6675	0.6946	0.0799	0.00503	0.00455	7
24	76.40%	1165	11	0.0162	0.7831	0.6403	0.00413	0.00481	0.00478	9
13	76.29%	995	12	0.0277	0.6345	0.6829	0.3999	0.00011	0.00567	6
18	76.29%	915	13	0.0121	0.6784	0.5958	0.1612	0.00491	0.1732	8

Top five of 150 Optuna trials, ranked by in-region validation accuracy (Acc.); trial 32 (bold) was selected for the field-level XGBoost. Est.: number of estimators; Depth: maximum tree depth; LR: learning rate; SubS: row subsample; ByTree: column subsample per tree; Gamma: minimum split-loss reduction; Regα/Regλ: L1/L2 regularization; MinCW: minimum child weight.

TabNet. At each of n_{steps} decision steps, an attentive transformer produces a sparse feature mask via *sparsemax*,

$$\begin{aligned} \mathbf{M}[i] &= \text{sparsemax}\big(\mathbf{P}[i-1] \odot h_i(\mathbf{a}[i-1])\big), \\ \mathbf{P}[i] &= \textstyle\prod_{j=1}^{i} \big(\gamma - \mathbf{M}[j]\big), \end{aligned} \tag{S2}$$

with relaxation $\gamma = 1.5$; the masked features are handled by a feature transformer. We use $n_d = n_a = 64$, 5 decision steps, batch size 1024, and ensemble five seeds by averaging softmax outputs. The sparsemax masks are the mechanism the main paper's L-TAE-S grafts onto a temporal-attention backbone (Arik and Pfister, 2021; Muruganantham et al., 2024).

L-TAE. Given an embedded sequence $\{\mathbf{z}_t\}_{t=1}^{T}$, a learned master query attends over time; for a single head

$$\mathbf{o} = \sum_{t=1}^{T} \alpha_t \, \mathbf{v}_t, \qquad \alpha_t = \frac{\exp(\mathbf{q}^\top \mathbf{k}_t / \sqrt{d})}{\sum_{t'} \exp(\mathbf{q}^\top \mathbf{k}_{t'} / \sqrt{d})}, \tag{S3}$$

with channel grouping keeping the encoder lightweight (Garnot and Landrieu, 2020). Evaluated at pixel and field level as a five-seed ensemble.

Transformer encoder. After the same band embedding and month-aware positional encoding as L-TAE, the sequence passes three pre-norm multi-head self-attention layers ($d_{\text{model}} = 128$, 8 heads, feed-forward width 256), is mean-pooled, and classified (Vaswani et al., 2017; Rußwurm and Körner, 2020). Unlike L-TAE-S it has no feature-sparsity mechanism, serving as the capacity-versus-sparsity control. Five-seed ensemble under the same weighted-focal objective.

TempCNN. Stacked 1D convolutions along the temporal axis with batch normalization and dropout, followed by a softmax head (Pelletier et al., 2019); pixel and field variants, five seeds.

Patch models. Each field is tiled into 100 m patches ($\sim 10 \times 10$ pixels) resampled to 128×128 (Fig. S5). The multi-channel 2D CNN treats each band×month as a channel ($C = 60$) through two convolutional blocks; the 3D CNN treats each patch as a spatiotemporal volume ($T = 10$, $C = 6$) through three 3D-convolutional blocks with a weighted focal loss (Fig. S6). Both are five-seed ensembles aggregated to a field label by majority vote.

S-B. Performance Metric Definitions

Supports main paper Sec. 3. Let N be the number of samples, C the classes, and n_c the support of class c. Cohen's Kappa is $\kappa = (p_o - p_e)/(1 - p_e)$ with p_o the observed agreement and $p_e = \sum_c (n_{c,\text{true}}/N)(n_{c,\text{pred}}/N)$. Macro-F1 (primary) and weighted F1 are

$$\mathrm{F1}_m = \frac{1}{C}\sum_{c=1}^{C}\mathrm{F1}_c, \quad \mathrm{F1}_{\mathrm{w}} = \sum_{c=1}^{C}\frac{n_c}{N}\,\mathrm{F1}_c, \tag{S4}$$

with $\mathrm{F1}_c = 2\,\mathrm{Prec}_c\,\mathrm{Rec}_c/(\mathrm{Prec}_c + \mathrm{Rec}_c)$. Cross-entropy (Xent) is the AI4FoodSecurity field-level scoring metric (AI4EO, 2021); because submissions are hard labels, it is evaluated on the one-hot predicted label clipped to $[\epsilon, 1-\epsilon]$, $\epsilon = 10^{-7}$:

$$\begin{aligned} \mathrm{Xent} &= -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\,\ln \hat{q}_{i,c}, \\ \hat{q}_{i,c} &= \mathrm{clip}\big(\nVdash[\hat{y}_i = c],\ \epsilon,\ 1-\epsilon\big). \end{aligned} \tag{S5}$$

S-C. Training Cost

Supports main paper Sec. 3 and the Discussion. Table S3 reports wall-clock training time for a single run of each model on one workstation (RTX 3080 Ti). The robust models that transfer best (the `xr_fresh` tree ensembles and the field-level L-TAE-S) are among the cheapest to train, while the most expensive (the patch CNNs and the pixel-level Transformer) do not lead the holdout ranking.

S-D. Ablation Studies

Supports main paper Sec. 5 ("Two Robustness Levers"). We attribute the holdout ranking primarily to two levers—built-in sparse feature selection and field-level aggregation—and show that ensemble size and the imbalance-handling objective are second-order: they shift absolute scores but never reorder the robust and fragile model groups.

(i) Field-level aggregation (main Table 2) recovers most of the transfer gap for dense temporal nets (L-TAE $-0.20 \to -0.07$; L-TAE-S $-0.17 \to -0.04$) but harms TabNet ($0.60 \to 0.43$). *(ii) Sparse selection versus attention capacity*: a plain Transformer with far greater capacity transfers no better than L-TAE (both ST $\mathrm{F1}_m = 0.58$), whereas grafting a sparse channel gate (L-TAE-S) lifts transfer to 0.60. *(iii) Resampling*: the *Stacking* and *SMOTE Stacked* ensembles differ only in SMOTETomek resampling; resampling widens the gap ($-0.11 \to -0.21$) and lowers holdout $\mathrm{F1}_m$ ($0.53 \to 0.49$).

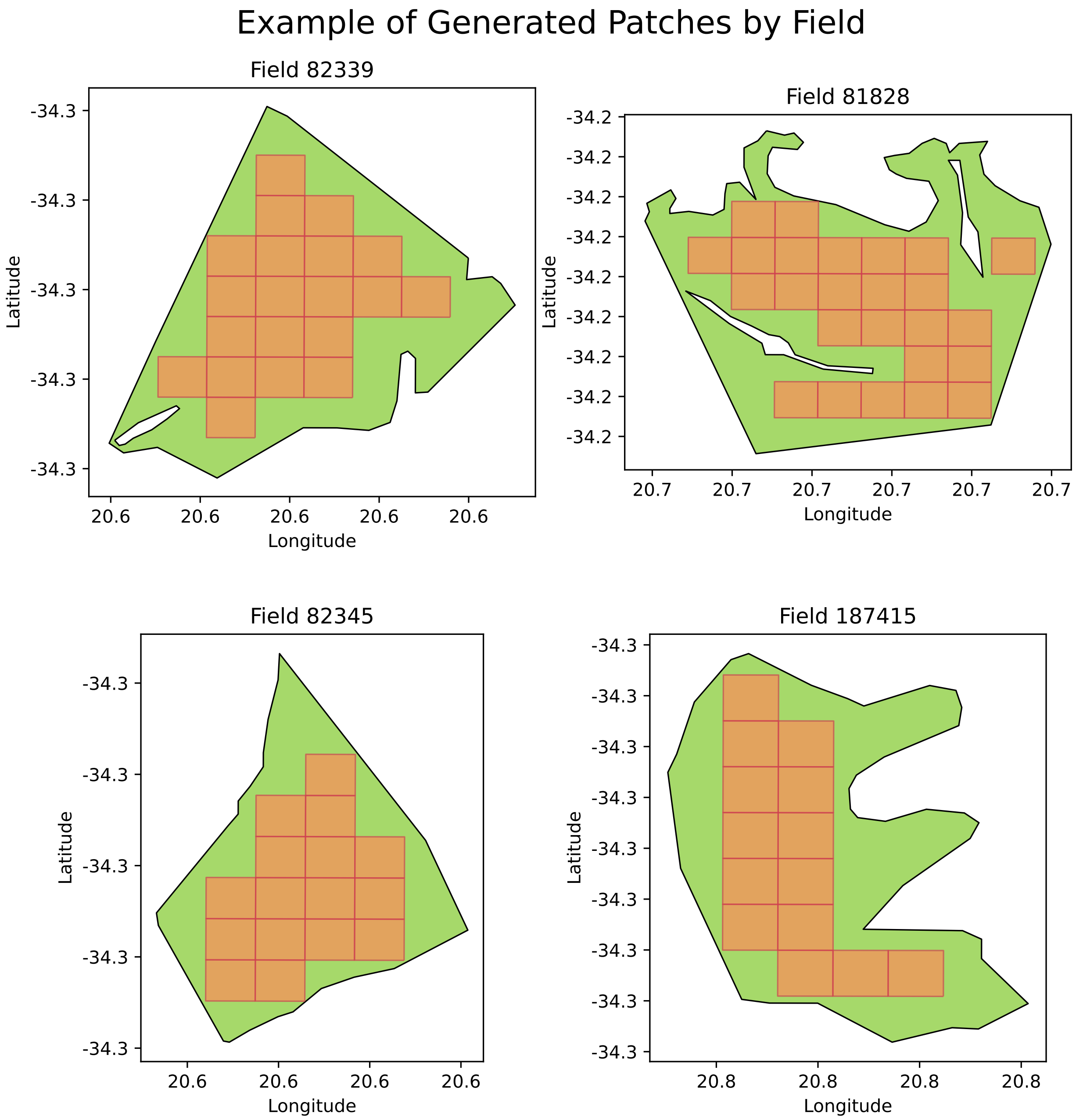


Figure S5: Patch generation per field. Each field is tiled into 100 m patches—only ∼10 × 10 pixels at the 10 m Sentinel-2 resolution—and resampled to 128 × 128. These patches are the input to the patch-based 2D and 3D CNNs, whose per-patch predictions are later aggregated to a field label by majority vote.

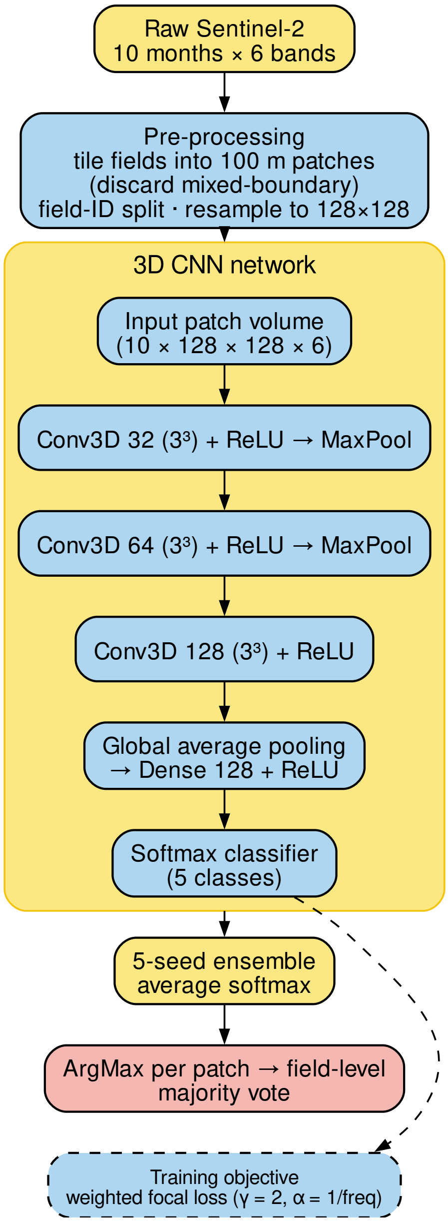


Figure S6: Patch-level 3D CNN pipeline: each field is tiled into 100 m patches resampled to 128×128; a five-seed ensemble averages softmax outputs, and per-patch predictions are aggregated to a field label by majority vote.

Table S3: Per-seed training time for each model.

Model	Level	Features	Time / run
Classical, field-level			
XGBoost	field	`xr_fresh`	12 s
LightGBM	field	`xr_fresh`	31 s
Voting	field	`xr_fresh`	109 s
Stacking	field	`xr_fresh`	109 s
SMOTE Stacked	field	`xr_fresh`	99 s
Classical, pixel-level			
XGBoost	pixel	`xr_fresh`	31 s
LightGBM	pixel	`xr_fresh`	31 s
Logistic Regression	pixel	`xr_fresh`	8.0 min
Random Forest	pixel	`xr_fresh`	73 min
Deep, field-level (raw sequences)			
TabNet (`xr_fresh`)	field	`xr_fresh`	34 s
TabNet (raw, field-avg)	field	raw	39 s
CNN-BiLSTM	field	raw	53 s
L-TAE	field	raw	61 s
TempCNN	field	raw	64 s
L-TAE-S	field	raw	18 s
Deep, pixel-level (raw sequences)			
CNN-BiLSTM	pixel	raw	10 min
L-TAE-S	pixel	raw	24 min
TempCNN	pixel	raw	29 min
L-TAE	pixel	raw	38 min
TabNet (raw)	pixel	raw	61 min‡
Transformer	pixel	raw	97 min
Deep, patch-level (raw sequences)			
3D CNN	patch	raw	22 min
Multi-Channel 2D CNN	patch	raw	38 min

Deep-model times are the mean per seed of a 5-seed ensemble (total wall-clock divided by five); classical models are a single best-configuration fit, excluding the one-time Optuna search. ‡Measured on the 75% training fraction, the best-transferring TabNet pixel configuration.

Table S4: Ablation (iv): spatial-transfer holdout performance versus seed-ensemble size.

Model	Ensemble	$\mathbf{F1}_m$	κ
TabNet (field)	1 seed	0.31 ± 0.02	0.18 ± 0.03
	3 seeds	0.39	0.29
	5 seeds	0.43	0.34
L-TAE-S (field)	1 seed	0.56 ± 0.02	0.47 ± 0.02
	3 seeds	0.58	0.50
	5 seeds	0.60	0.52
CNN-BiLSTM (field)	1 seed	0.48 ± 0.02	0.31 ± 0.02
	3 seeds	0.52	0.34
	5 seeds	0.51	0.33

By re-inference of saved per-seed checkpoints. Single-seed rows report mean±s.d. across five seeds.

(iv) Ensemble size (Table S4): seed-ensembling gives a small, monotone lift but does not change the ordering—a single-seed L-TAE-S ($\text{F1}_m = 0.56$) already exceeds the five-seed CNN-BiLSTM and TabNet field ensembles. *(v) Imbalance-handling objective* (Table S5): the robust L-TAE-S is almost invariant across five loss/sampler settings (0.59–0.60), whereas the fragile CNN-BiLSTM is sensitive (0.41–0.56); even CNN-BiLSTM's best setting (0.56) stays below L-TAE-S's worst (0.59), so the objective never reorders the two.

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Table S5: Ablation (v): effect of the imbalance-handling objective and resampling on spatial-transfer holdout performance.

Model	**Loss**	**Resample**	$\mathbf{F1}_m$	κ
TabNet (field)	Weighted focal (γ=2)	no	0.43	0.34
	Weighted CE (γ=0)	no	0.42	0.33
	Plain CE	no	0.35	0.25
	Weighted focal (γ=2)	yes	0.40	0.23
	Plain CE	yes	0.46	0.37
L-TAE-S (field)	Weighted focal (γ=2)	no	0.60	0.52
	Weighted CE (γ=0)	no	0.60	0.50
	Plain CE	no	0.59	0.50
	Weighted focal (γ=2)	yes	0.60	0.49
	Plain CE	yes	0.60	0.50
CNN-BiLSTM (field)	Weighted focal (γ=2)	no	0.51	0.34
	Weighted CE (γ=0)	no	0.49	0.34
	Plain CE	no	0.41	0.30
	Weighted focal (γ=2)	yes	0.56	0.39
	Plain CE	yes	0.50	0.34

Field-level, 5-seed ensembles. "Resample" denotes class-balanced oversampling.

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